

# VTY Configuration



## Telnet Configuration

**Step 1 ) Modify Telnet Server/Client**

**Switch(config)# interface vlan** *vlan-id*
**Switch(config-if)# no shutdown**
**Switch(config-if)# ip address** *ip-address subnet-mask*

### Example

**Switch(config)# interface vlan** *1*
**Switch(config-if)# no shutdown**
**Switch(config-if)# ip address** *192.168.1.1 255.255.255.0*



## Telnet Configuration

### Step 2 ) Device needs an IP address

## LAN Switch Interface Status Codes

| Line Status | Protocol Status | Interface Status | Typical Root Cause |
|---|---|---|---|
| Administratively Down | Down | disabled | The interface is configured with the **shutdown** command. |
| Down | Down | notconnect | No cable; bad cable; wrong cable pinouts; the speeds are mismatched on the two connected devices; the device on the other end of the cable is powered off or the other interface is **shutdown**. |
| Up | Down | notconnect | An interface up/down state is not expected on LAN switch interfaces. |
| Down | down (err-disabled) | err-disabled | Port security has disabled the interface. |
| Up | Up | connected | The interface is working. |



## Telnet Configuration

Step 2 ) Device needs an IP address

```
Switch (config)# interface type mod/num
Switch (config-if)# no shutdown
Switch (config-if)# ip address ip-address subnet-mask
```

```
Switch (config)# interface fastEthernet 0/1
Switch (config-if)# no shutdown
Switch (config-if)# ip address 192.168.1.1 255.255.255.0
```



## Telnet Configuration

**Step 3 ) Providing Security for the User mode**

**Cisco(config)#Line VTY** *first-line last-line*
**Cisco(config-line)#Password** *password*
**Cisco(config-line)#Login**

Example

**Cisco(config)#Line VTY** *0 15*
**Cisco(config-line)#Password** *test*
**Cisco(config-line)#Login**



## Telnet Configuration

**Step 3 ) Providing Security for the User mode**

**Cisco(config)#Username** *username* **password** *password*
**Cisco(config)#Line VTY** *first-line last-line*
**Cisco(config-line)#Login Local**

Example

**Cisco(config)#Username** ***Beygi*** **password** *test*
**Cisco(config)#Line VTY** *0 15*
**Cisco(config-line)#Login Local**



## Telnet Configuration

**Step 4 ) Providing Security for the Privilege mode**

**Cisco(config)#enable password** *password*
**Cisco(config)#enable secret** *password*



## Telnet Configuration

### Switch sample configuration

**Switch(config)# interface Vlan** *1*
**Switch(config-if)# no shutdown**
**Switch(config-if)# ip address** *192.168.3.235 255.255.255.0*
**Switch(config)# Line vty** *0 15*
**Switch(config-if)# Pasword** *cisco*
**Switch(config-if)#Login**
**Switch(config)#Enable password** *cisco*



## SSH Configuration

**Step 1 ) Do all the things you need to do to configure Telnet**

**Note:** User mode should be protected by Username/Password



## SSH Configuration

**Step 2 ) Modify SSH Server/Client**

**Note :** Device IOS must support security features for enabling SSH.

**Enable SSH Server**

**Cisco(config)#Line VTY** *first-line last-line*
**Cisco(config-line)#Transport input [Telnet|SSH|All|None]**

Example

**Cisco(config)#Line VTY** *0 15*
**Cisco(config-line)#Transport input All**



## SSH Configuration

Step 3 ) Key configuration

**Switch(config)# hostname** *hostname*
**Switch(config)# ip domain-name** *domain-name*
**Switch(config)# Crypto key generate RSA**

**Switch# show crypto key mypublickey rsa**

Example

**Switch(config)# hostname** ***Sadjad***
***Sadjad*** **(config)# ip domain-name** ***Sadjad.local***
***Sadjad*** **(config)# Crypto key generate RSA**